

Cosmic Star Formation History from HSC-CIB Tomography

Nathan Durrant¹, Tom Cornish², David Alonso³



¹Department of Physics, University of Warwick

²Department of Physics, Imperial College London

³Department of Astrophysics, University of Oxford



Introduction

The cosmic infrared background (CIB) is diffuse, ubiquitous radiation which originates from dust in star-forming galaxies across the Universe. By cross-correlating the CIB with the positions of galaxies from different epochs (a technique known as *tomography*), we can trace the star formation activity in the Universe across cosmic time, providing insight into how galaxies form and evolve throughout cosmic history. Here we perform a tomographic analysis combining the CIB with a novel galaxy sample to place new constraints on the star formation history of the Universe.

Data

Our analysis uses two cosmic tracers: *galaxy clustering* (GC) and the CIB. We take this data from the Hyper Suprime-Cam Subaru Strategic Program (HSC-SSP) Public Data Release 3 and the 857 GHz *Planck* CIB maps [1], respectively.

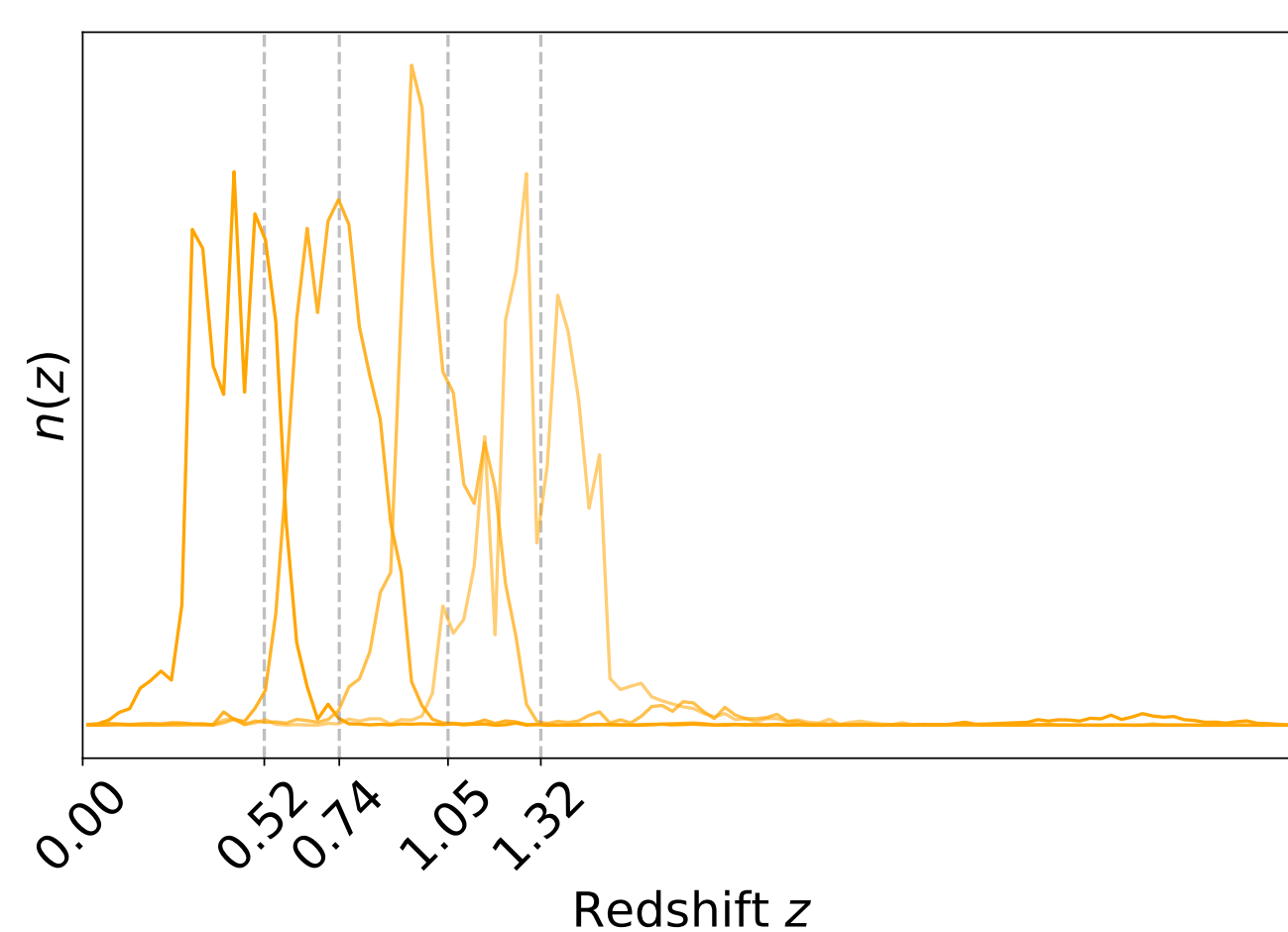


Figure 1: The redshift distribution for the binned galaxy clustering data. The grey dashed lines signify the weighted mean for each bin, with the values included along the z -axis.

Angular Power Spectra

These are our choice of *observable*. These C_ℓ s measure the variance of a cosmic tracer across the sky, as a function of angular scale. A higher multipole ℓ corresponds to a lower angular scale.

Using these we obtain an auto-correlation power spectrum for each GC redshift bin (see Figure 1) as well as a cross-correlation between each GC bin and the CIB map.

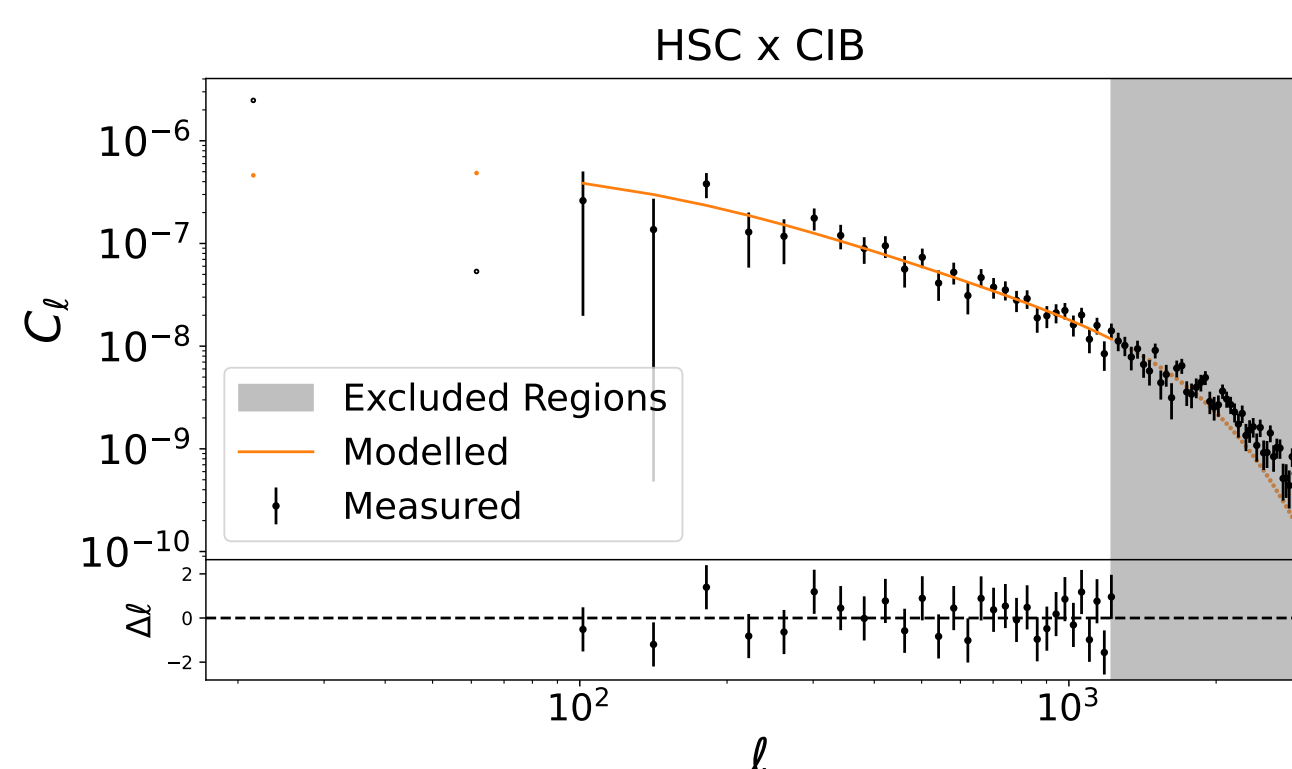


Figure 2: HSCx CIB cross-correlation spectrum for the final HSC bin. The orange line represents the best fit given by our modelling.

Measuring $\langle b\rho_{\text{SFR}} \rangle$

Using both power spectra, we employ MCMC sampling to jointly constrain, for each redshift bin, the galaxy bias b_g and the *bias-weighted star formation rate density* $\langle b\rho_{\text{SFR}} \rangle$.

We employ the following model [2] for the auto- and cross-correlation data, respectively:

$$\begin{aligned}\hat{C}_\ell^{gg} &= b_g^2 \tilde{C}_\ell^{mm,gg} + A_{gg} \tilde{C}_\ell^{mmk^2,gg} + N_{gg}, \\ \hat{C}_\ell^{g\nu} &= b_g \langle b\rho_{\text{SFR}} \rangle \tilde{C}_\ell^{mm,g\nu} + A_{g\nu} \tilde{C}_\ell^{mmk^2,g\nu} + N_{g\nu}\end{aligned}$$

The nuisance terms $\{A_{gg}, A_{g\nu}, N_{gg}, N_{g\nu}\}$ can be analytically marginalised, reducing the dimensionality of the parameter space from 24 to 8 and substantially reducing the chain convergence time. We apply scale cuts to avoid both contamination by observational systematics and the small-scale regime in which our model assumptions break down. These are indicated by the grey regions in Figure 2.

The gg spectrum independently constrains b_g , breaking parameter degeneracy; this is then used in the $g\nu$ spectrum to constrain $\langle b\rho_{\text{SFR}} \rangle$. The process yields four novel measurements, presented alongside existing values in Figure 3.

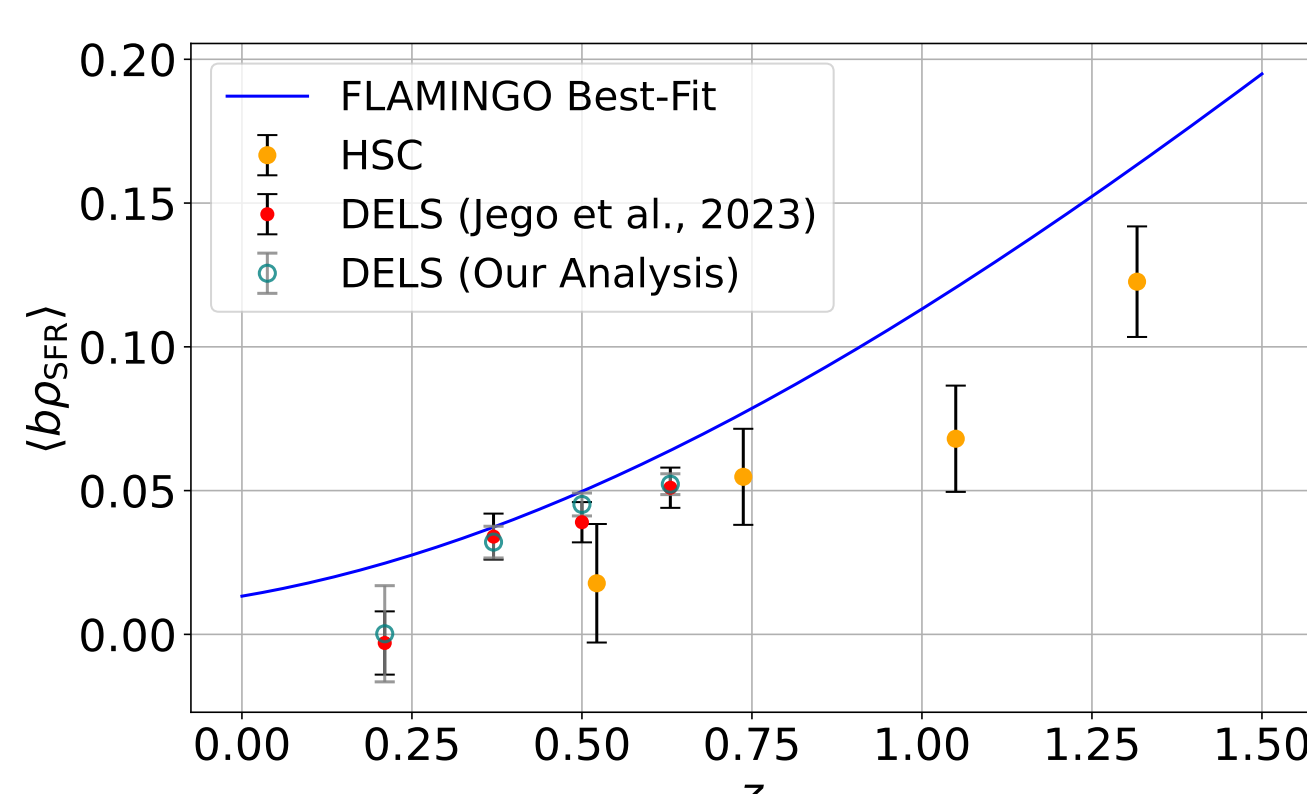


Figure 3: Measurements of $\langle b\rho_{\text{SFR}} \rangle$. The blue line represents the FLAMINGO predictions from [2]

Conclusions

We have presented four new measurements of $\langle b\rho_{\text{SFR}} \rangle$, across redshift, that can be further used in studies of SFR densities and constraining halo-model parameter space. This work demonstrates the efficacy of CIB tomography alongside the model from [2].

Our results, documented primarily in Figure 3, lie below the FLAMINGO best fit. This is suggestive of the effect of certain systematics, most probably:

- Masking: Current quality control flags may fail to remove all galaxies impacted by bright stars and artefacts, skewing density measurements.
- Redshift calibration: The modelling depends on redshift catalogues of known galaxies – this might only be a small subset and thus not representative.
- Survey properties and sample selection.

We are actively working to mitigate the effect of these.

Our pipeline produces values in close agreement to those found in [3] when performed on identical data. This gives us confidence that the under-prediction is data-driven rather than a computationally caused error and validates our methodology. This highlights the need for further work on mitigation of systematics in HSC tomographic studies.

Future Work

A natural extension to this work is to use a halo-based star formation rate model to remove the bias weighting from $\langle b\rho_{\text{SFR}} \rangle$. This gives a history of ρ_{SFR} . Such a process is well documented in [3] and we will follow the model formalism presented in [4].

References

- [1] Lenz et al. *ApJ*, 883(1):75, September 2019.
- [2] Maleubre et al. *MNRAS*, 545(2):staf2125, January 2026.
- [3] Jego et al. *MNRAS*, 520(2):1895–1912, January 2023.
- [4] Maniyar et al. *A&A*, 645:A40, January 2021.